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$$\begin{aligned} & \lim_{x \rightarrow \infty} \sqrt{25x^2 - 6} - 5x \\ & \lim_{x \rightarrow -\infty} \sqrt{25x^2 - 6} - 5x = +\infty + \infty = +\infty \\ & \lim_{x \rightarrow +\infty} \sqrt{25x^2 - 6} - 5x = +\infty - \infty \\ & = \lim_{x \rightarrow +\infty} \frac{(\sqrt{25x^2 - 6} - 5x)(\sqrt{25x^2 - 6} + 5x)}{\sqrt{25x^2 - 6} + 5x} \\ & = \lim_{x \rightarrow +\infty} \frac{25x^2 - 6 - (-5x)^2}{\sqrt{25x^2 - 6} + 5x} = \frac{25x^2 - 6 - 25x^2}{\sqrt{25x^2 - 6} + 5x} \\ & = \lim_{x \rightarrow +\infty} \frac{-6}{\sqrt{25x^2 - 6} + 5x} = 0 \end{aligned}$$

## References